

Nowcasting European LNG send-out from free public data: measured revision processes, calibrated filtering, and the absence of a daily trading edge

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Abstract

Daily gas send-out from north-west European LNG terminals is a direct short-term driver of the TTF and NBP markets, and commercial vendors sell reconstructions of it from proprietary satellite ship-tracking. We rebuild the signal entirely from free public sources and add the two layers vendors do not publish: a measurement of how the official data itself is revised, and calibrated uncertainty validated against later-published truth. A semi-Markov, Rao-Blackwellized particle filter delivers evening nowcasts $3\text{--}7\times$ more accurate than any daily baseline, with distribution-free coverage guarantees, and generalizes through the 2022–23 energy crisis. En route we document several unpublished properties of the official data: the platform’s intraday “estimate” appears to be a carry-forward of the previous day’s value; published nominations forecast worse than persistence; one gas day’s official record permanently contradicts itself by a full cargo. Intraday deviations follow a near-degenerate fractional bridge ($\hat{H} \approx 0.98$): each day is essentially one random ramp. Cargo arrivals extracted from tank mass balance match vessel-class physics to within one percent and are emphatically non-Poisson. Finally, we backtest the signal set against front-month TTF under a pre-registered protocol with costs, roll hygiene, and selection accounting, and report a comprehensive negative: across 6.5 years, no daily-frequency directional edge exists in any of these physical signals — including supply-break events, whose apparent edge inverts once crisis years are included, because at the daily horizon causality runs from price to flows as much as the reverse. The residual economic value of physical nowcasting concentrates at sub-daily horizons and in calibrated uncertainty, which is precisely where professional market access begins.

1 Introduction

The quantity of liquefied natural gas regasified and sent into the north-west European grid each day — “send-out” — is one of the few physical variables that moves the continent’s benchmark gas prices at short horizons. It is also, nominally, public: transmission operators, terminal operators, and the European transparency platforms all publish versions of it. In practice the publications are slow, mutually inconsistent, revised after the fact, and scattered across five differently-shaped interfaces, which is why a commercial industry (Kpler, Vortexa, ICIS) sells faster reconstructions built on proprietary satellite ship-tracking.

This paper asks what can be recovered from the free layer alone, and deliberately measures the parts the commercial products do not publish. Our contributions:

1. **An open reconstruction with measured plumbing.** We fuse five free sources (Section 2) and document their true timing and failure modes, several of which contradict the platforms’ own documentation: publication leads of ten hours between sources measuring the same quantity, hourly feeds trailing real time by two hours, silent all-null series, and API queries that hang indefinitely.
2. **The revision processes of the official record** (Section 3). Using retroactive version histories where they exist and a thrice-daily snapshot archive where they do not, we quantify how “estimated” becomes “confirmed”: the United Kingdom’s three-rung maturity ladder collapses to two informative rungs; the final-reconciliation rung never differs from its predecessor across 2,829 matched days; corrections have a heavy tail reaching 662.7 GWh; and one gas day’s official record permanently contradicts itself by a full cargo.
3. **A calibrated nowcaster** (Sections 5–6). A semi-Markov switching model with explicit-duration regimes, marginalized (Rao-Blackwellized) over its conditionally linear-Gaussian states, driven by a renewal arrival process calibrated on 1,446 extracted cargo discharges, observed through a fractional-bridge intraday channel, and wrapped in adaptive conformal calibration. Evening nowcasts achieve 3–7 $\times$ the accuracy of persistence with bands that empirically cover at their nominal rates, evaluated walk-forward over 1,552 days per terminal including the 2022–23 crisis.
4. **A comprehensive, disciplined negative** (Section 8). Under a pre-registered backtest protocol — information-aligned execution, transaction costs, roll hygiene, Newey–West inference, explicit selection accounting, and a frozen 2026 holdout — no daily-frequency directional edge on front-month TTF survives in any of the signals, their changes, or their tails. We bound the price response of ordinary flow surprises, show the market’s standard stress dial (storage fullness) is itself a poor directional signal, and explain the sign inversion of supply-break trading by reverse causality at crisis peaks.

Three literatures intersect here without quite covering it. Vessel-arrival prediction is dominated by statistical and machine-learning taxonomies with no stochastic-process formulations [6]; the maritime inventory-routing literature models exactly our constraint set but prescriptively, with import-terminal inventories under-modelled and uncertainty handling described as in its infancy [8]; and the LNG economics literature establishes that inter-regional arbitrage is structurally damped [9, 10], which our market-response measurements corroborate from the physical side.

2 Data: five free sources and their measured behaviour

Table 1 summarizes the sources and what live measurement revealed about each. Three findings deserve emphasis because they contradict reasonable expectations. First, the ENTSOG transparency platform publishes confirmed daily flows at $\sim$ 09:20 CET the next morning — ten hours *before* the official LNG platform’s 19:30 publication of the same gas day, and its hourly series trails real time by only two hours, so an intraday nowcast can lead the official print by half a day using nothing exotic. Second, entire series can be silently absent: the UK terminals stopped reporting to the European LNG platform before 2024 (974 consecutive days of null status), and one terminal’s hourly series consists entirely of rows whose values are null — a naive aggregation reads it as zero flow. Third, the platforms’ own forecasts are uninformative:

Source	Access	Cadence	Measured behaviour
ENTSOG transparency	keyless	daily/hourly	day D confirmed $\sim 09:20$ $D+1$; hourly ~ 2 h lag; nominations for D at $\sim 16:00$ $D-1$ but uninformative; UK rows backfilled ~ 6 days; LNG-point history begins 2021-04; some queries hang with zero bytes
GIE ALSI/AGSI	free key	daily	19:30 + 23:00 publications; dual-unit inventories; UK absent since ≥ 2024 ; intraday estimate row equals the prior day’s confirmed value (carry-forward; preliminary)
National Gas portal	keyless	daily + 2-min	$D+1$ physical at 12:01; $D+2/M+15$ ladder; full retroactive version history; per-sub-terminal 2-minutely flows, not archived by the operator; history floors at 2021-08
aisstream.io (AIS)	free key	streaming	terrestrial only (no mid-ocean); berth calls resolvable; receiver gaps at specific ports; destination fields $\sim 52\%$ erroneous [7]
Prices	open	daily	UK SAP (portal); TTF front-month (2017–) for analysis only

Table 1: Sources and live-measured behaviour (measurements: 2026-09-01/02).

archived nominations forecast realized send-out worse than persistence at every terminal tested (Table 2), with roughly 70% of archived values being post-hoc renominations.

3 How the official record revises itself

The United Kingdom publishes each day’s send-out at three nominal maturities, and — unusually — serves every historically published version of every value, so its revision process is reconstructible retroactively. Across 17,628 published versions (2024–2026): 27.8% of values are republished, but the median republication changes nothing — the $D+2$ commercial series is systematically re-issued about thirty days later, almost always bit-identical. The informative structure is a heavy tail: send-out corrections reach 662.7 GWh and tank-stock corrections 340 GWh (the latter arriving the next day). The ladder’s final rung is empty of information: the “ $M+15$ ” items in fact publish at about $M+1$ and never differ from the final $D+2$ value (maximum discrepancy 0.00 GWh over 2,829 matched days). The next-morning physical figure is within 0.02 GWh of final at the 99th percentile — so revision risk is a rare-event phenomenon, not a diffuse noise floor.

The canonical illustration is Grain, gas day 2025-04-21: the physical $D+1$ series and the independent ENTSOG measurement both record ~ 662 GWh — near maximal send-out — while the commercial series, having initially agreed, was retracted to exactly zero a month later and fi-

nalized there. The official record permanently disagrees with itself by a full cargo, and that day’s D+1 also published three days late: publication timing has tails too. Any pipeline that silently trusts one official series inherits such events, which is the concrete argument for heavy-tailed observation models and multi-source triangulation (our cross-validation of the two independent UK measurements agrees to 0.2–0.4% of flow on ordinary days).

On the European side no version history exists, so we archive the platforms ourselves three times daily. A preliminary finding from the young archive: the intraday “estimated” row for the running gas day equals the previous day’s confirmed value, digit for digit, at all four terminals sampled — i.e. the official intraday estimate appears to be a carry-forward containing no information. We flag this as preliminary pending a longer archive.

4 Cargo arrivals from mass balance

Tank inventories close the mass balance: $A_t = \Delta I_t + S_t$ up to boil-off. Extracting arrival events this way over 2024–2026 yields 1,446 discharges. Their sizes constitute an independent test of the whole measurement chain, because discharge energy is separately predicted by vessel physics (class capacity $\times$ loaded fill $\times$ voyage boil-off $\times$ retained heel $\times$ calorific content):

	median event (GWh)	172k-class prediction	deviation
Dunkerque	1095	1096	−0.1%
Gate	1105	1096	+0.8%
Zeebrugge	1096	1096	0.0%
Montoir	1092	1096	−0.4%

Four terminals’ medians match independent physics to within one percent. Attribution caveats are themselves informative: terminals with multiple berths chain overlapping ships into multi-day clusters, and small-tank terminals stage single cargoes across pauses (close pairs at Dragon sum to Q-Flex energies, e.g. 1351.5 vs. the 1357 class value).

The arrival *process* is emphatically non-Poisson. The empirical hazard of an arrival at a one-day gap is exactly zero at every terminal — a berth-turnaround refractory period no memoryless model can produce — and rises with gap age at scheduled terminals (Gate reaches 0.56 by day eight). The filter therefore drives arrivals with a Negative-Binomial renewal hazard on observed days-since-last-cargo rather than a constant rate. AIS draught inversion, the folkloric route to cargo size, is strictly dominated where tank data exist: ballast compensation hides roughly 60% of the mass change, leaving $\pm 38\%$ errors against mass balance’s $\sim 1\%$; draught earns its keep instead as a laden/ballast classifier for ships in transit.

5 A semi-Markov, Rao-Blackwellized nowcaster

States and regimes. Let S_t be send-out and I_t tank inventory (GWh) on gas day t . Send-out is a decision process with metastable states: an *idle* regime with $S_t = 0$ exactly, and an *active* regime. Regime r_t carries its age a_t , and switches with the discrete hazard $\lambda_r(a)$ of a Negative-Binomial sojourn law fitted to observed spell lengths — the explicit-duration (semi-Markov) structure. Sojourn laws are first-passage-time distributions of the underlying decision process; their strongly non-geometric shape is why constant-hazard Markov switching misfits long idle spells.

Intraday deviations are one persistent ramp, not Brownian noise

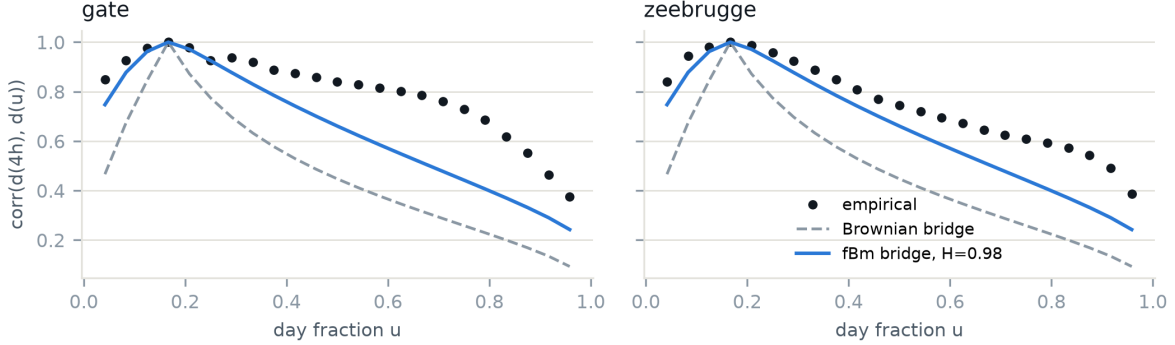


Figure 1: Correlation of intraday deviations with the hour-4 value: empirical structure (dots) against the Brownian bridge and a fractional bridge with $H = 0.98$. Deviation paths are persistent ramps, not noise.

Conditionally linear-Gaussian dynamics. Given the discrete path (regimes, ages, innovation-component indicators z_t , latent Student- t scales w_t , and sampled arrival inputs A_t), the continuous dynamics are linear-Gaussian:

$$\begin{aligned} S_t &= S_{t-1} + \eta_t, & \eta_t \mid z_t, w_t &\sim \mathcal{N}(0, \sigma_{z_t}^2 w_t) \quad (\text{active, staying}), \\ S_t &\sim \mathcal{N}(\mu_r, \sigma_r^2) \quad (\text{active, entering}); & S_t &= 0 \quad (\text{idle}), \\ I_t &= (1 - b) I_{t-1} + A_t - S_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \sigma_I^2). \end{aligned}$$

Each particle therefore carries only discrete variables plus Kalman sufficient statistics (m_t, P_t) for (I_t, S_t) ; the continuous states are integrated out exactly (the marginalized particle filter of [3]), and particle weights multiply exact Gaussian predictive likelihoods. The idle regime is the $S = 0$ boundary, which dissolves the truncation problem where it matters most. In practice 500 particles match a 4,000-particle bootstrap filter intraday at fifteen times the speed, and the construction yields a byproduct no bootstrap filter provides cleanly: a daily posterior probability that the terminal is idle. One implementation subtlety deserves record: a posterior *view* at an intraday information set must apply the Kalman update to every particle’s mean, not merely reweight — reweighting alone silently destroys the intraday horizons.

The intraday observation channel. Cumulative metered flow through day-fraction u satisfies $C(1) = S$, so the deviation $d(u) = C(u)/S - u$ is bridge-like: $d(0) = d(1) = 0$. The natural first model is a Brownian bridge, $\text{Var}[d(u)] \propto u(1 - u)$. The data reject it in an instructive direction: across 789–947 active days per terminal, the empirical correlation $\text{corr}(d(4h), d(11h))$ is 0.71–0.86 against the Brownian bridge’s 0.49, and a fractional Brownian bridge fits with $\hat{H} \approx 0.98$ at every terminal, reducing correlation misfit by 81–87% (Figure 1). $H \rightarrow 1$ is the degenerate limit in which each day’s deviation is a single random ramp — operationally, the morning reveals most of the day. The filter uses the bridge form with terminal-specific mean profiles $m(u)$ (Zeebrugge ramps +2.4% by hour eleven) and robust per-hour scales fitted on the training window only; the fitted kernel licenses updating at arbitrary intraday times.

Terminal	H0	H1 (midday)	H2 (evening)	persistence	nominations	cov90
Gate	20.0	9.1	5.2	20.5	71.4	0.91
Zeebrugge	46.5	28.3	22.6	46.3	123.2	0.85
Eems	18.7	—	—	18.5	36.0	0.91
Dunkerque	37.2	—	—	37.0	224.2	0.90
Grain	27.2	—	—	27.1	—	0.90
South Hook	22.5	—	—	22.2	—	0.91

Table 2: MAE (GWh/day), walk-forward 2022-06 – 2025-12 (1,552 days per terminal; Eems from its 2022 commissioning); cov90 is conformal coverage at nominal 0.90. Terminals without a usable public intraday feed run H0 only. On the shorter post-crisis window (2024-06 – 2026-08) the evening figures sharpen further (Gate 4.5, Eems 2.1).

Calibration with guarantees. Model-based bands carry interpretation; an adaptive conformal layer [1, 2] carries the guarantee. For each stream we track the empirical quantile of normalized errors by a Robbins–Monro recursion and report the adjusted interval; long-run coverage at every nominal level then holds with no assumptions on the data-generating process (the guarantee is marginal over time, not conditional — stated plainly). Walk-forward coverage lands at 0.47–0.61 against a nominal 0.50 and 0.85–0.93 against 0.90 across all terminals, horizons, and eras.

Parameter learning. Reference parameters are transparent moment fits; particle Gibbs with ancestor sampling [4] over the semi-Markov trajectory provides full posteriors in seconds per terminal. Its chief empirical lessons were about identifiability rather than point estimates: blocks the data cannot see (idle sojourns at a terminal that never idles) require tethering priors or the sampler random-walks; mixture components label-drift on long histories without an ordering constraint; and one restart-level posterior moved by a factor of two against its moment fit. Parameter uncertainty is otherwise modest and the filter’s evaluation is insensitive to the refit.

6 Walk-forward evaluation

Every fit uses only data preceding the evaluation day; terminals commissioned mid-sample are fitted on their first year and evaluated after. Table 2 reports the extended window — 1,552 days per terminal from June 2022, spanning the crisis — for the bootstrap reference filter with bridge observations and conformal bands.

Three structural readings. At the day-ahead horizon the filter ties persistence *everywhere* — correctly, because the only public day-ahead instrument (nominations) forecasts worse than persistence and the fitted blend weight rejects it; genuine day-ahead information would have to come from arrival pipelines and economics, which Section 8 prices. Where an intraday feed exists, the evening nowcast is 2–4× (post-crisis: 3–7×) more accurate than any daily baseline, half a day before the official print (Figure 2). And the uncertainty is honest: bands widen exactly at regime breaks, and conformal coverage holds through the crisis era.



Figure 2: Gate, June–August 2026: evening nowcast with 90% band against the figure published the next morning. The band flares precisely at the outage cliffs.

7 A physical stress index, pre-registered for winter

Tanks plus burn rate define *days of cover* — how long current send-out is physically sustainable without new cargoes. In-sample (2024–2026), adding cover to the market’s standard dial (storage fullness) raises the AUC for predicting rare TTF jumps within 7–14 days from 0.77–0.80 to 0.85, and for UK price spikes from 0.62 to 0.73. We deliberately do not claim this: thresholds are in-sample, windows overlap, only two winters exist, and two components show seasonal-confounding signatures. The out-of-sample test is pre-registered and imminent: Europe entered September 2026 with storage at 65% against a $\sim 90\%$ norm (Figure 3), and the index will be scored live through winter 2026–27 with the archive as timestamped evidence.

8 Economic content: a comprehensive negative

Does any of this move markets, and can it be traded? We answer in three escalating tests, and the answer is a bounded, mechanism-consistent no at the daily horizon.

Ordinary surprises are not priced. Over 564 trading days, a one-sigma (≈ 109 GWh) aggregate send-out surprise moves front-month TTF by at most 0.16% the same day, $|t| < 1$ under heteroskedasticity-robust inference — noise against 3–5% daily volatility. Typical daily flow news is 1–3% of regional demand and the front month ignores it.

The backtest protocol. Positions on front-month TTF at the daily close, from signals verifiably published before it; five-basis-point costs per unit turnover; continuous-contract roll windows excluded; online z-scoring with no lookahead; Newey–West t -statistics; every strategy variant counted against selection (eight tried); and a holdout (2026) frozen until the strategy set is final. Over 2018–2025 (1,642 tradable days, $\text{se}(\text{SR}) \approx 0.39$):

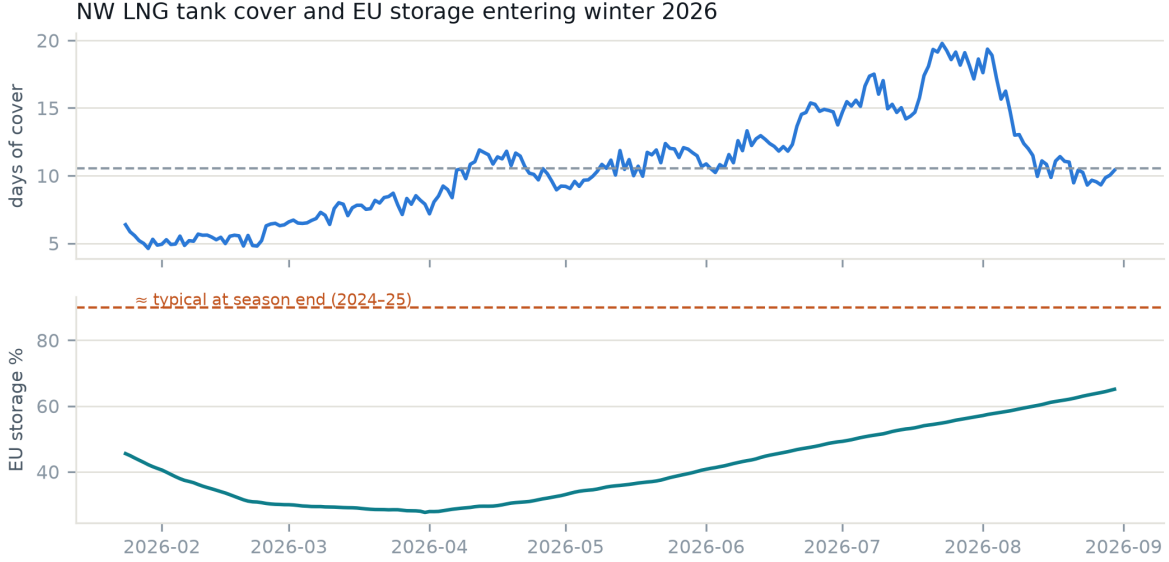


Figure 3: Days-of-cover and EU storage fullness entering winter 2026.

strategy	SR (net)	t_{NW}	maxDD	active
arrival drought	0.82	2.16	95%	100%
12-day momentum	0.77	2.03	88%	100%
Δcover (14d)	0.19	0.53	116%	100%
cover level	-0.18	-0.45	184%	100%
long-only	-0.23	-0.58	306%	100%
storage fullness	-0.66	-1.66	363%	100%
flow news (continuous)	-0.69	-1.80	151%	41%
supply-break events	-0.81	-2.06	92%	4%

The discipline earned its keep twice. First, the development window’s best strategy (Δcover , SR 1.50 on eighteen months) collapsed to 0.19 on full history — selection luck, caught before the holdout. Second, the last live hope — trading supply-break events detected intraday — grew from 18 to 65 trades when the evaluation was extended into the crisis years and returned SR -0.81 : not zero but *wrong-signed*, with a coherent mechanism. At the daily horizon, causality runs from price to flows as much as the reverse: extreme prices themselves curtail send-out (diversions, slot cancellations), so supply-loss events cluster near price peaks, and buying them loses. The best surviving row is statistically indistinguishable from the plain momentum beside it over a window containing the 2021–22 rally, and neither clears best-of-eight selection. The market’s standard stress dial, storage fullness, is itself a poor directional signal over 6.5 years. The holdout remains frozen; nothing qualifies to touch it.

Interpretation. These results are jointly consistent with the damped-arbitrage structure of the LNG market [9, 10] and with daily closes efficiently impounding public physical information. What the nowcaster demonstrably supplies — accuracy inside the day and calibrated uncertainty at breaks — is valuable precisely at horizons and in instruments (prompt, within-day, volatility) that free daily data cannot test and retail access cannot reach. We state the monetization

honestly: this is a measurement instrument, not a trading system.

9 Limitations

No free historical satellite AIS exists, so the ship-pipeline layer of the stress index accrues forward-only and is unvalidated. The intraday Estimated-row finding rests on a young archive. Zeebrugge’s transshipment traffic leaves residual miscalibration at its intraday horizons (raw coverage 0.65 at H2 in the crisis era; conformal repairs to 0.85). Interquartile bands run somewhat wide. The stress index’s positive in-sample margin awaits its pre-registered winter. History floors bind: ENTSG’s LNG series begin 2021, the UK portal’s in August 2021. And the back-test speaks only to daily-frequency, front-month, long/short directional trading — the negative is broad but not universal.

Reproducibility

All code, snapshot archives, and each table’s generating script: <https://github.com/solwilliams6305/LNG-send-out-nowcasting> (**[pin commit at submission]**). Data are credited per source terms (GIE ALSI/AGSI; ENTSG; National Gas; aisstream.io); TTF prices were used for analysis only and are not redistributed. A live dashboard rebuilt from the same pipeline runs at the repository’s Pages site.

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